

VEER ARORA

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PROFILE

Backend and AI engineer building systems that check their own output: reconciliation gates that test a parse against the source's own arithmetic, policy gates on agent actions, retrieval that cites the passage it used, and evaluation harnesses with held-out splits. At L&T Finance, validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow, ahead of its production release. Recent projects include a bank-statement ledger that proves it balances to the paisa before it categorises anything, an LLM support agent whose every write passes a policy and escalation gate, and a RAG system whose recall@5 rose from 0.37 to 0.77 against a measured ceiling of 0.85. B.Tech Computer Science & Design, 2026.

SKILLS

AI & LLM LLM tool-calling agents, RAG, LangGraph, evaluation harnesses (tool-selection, task-success, grounding, cost and latency), held-out test splits, OpenAI, Anthropic, Google Gemini and Groq APIs

Backend & Web FastAPI, PostgreSQL, row-level security, SQLAlchemy, Alembic, REST APIs, GitHub Actions CI, pytest, Docker, Git, React, TypeScript

Languages & Data Python, SQL, DuckDB, dbt, Pandas, NumPy, data cleaning and validation, EDA

Documents & Extraction PDF table recovery with pdfplumber (ruled and borderless), encrypted PDFs, scanned-vs-digital triage and OCR, ReportLab PDF generation, openpyxl

Cloud & MLOps Google Cloud Platform, Kubeflow, pipeline testing, UAT

Analytics & ML correlation analysis, descriptive statistics, outlier and anomaly detection, scikit-learn

Reporting & Tools Power BI, Matplotlib, Seaborn, Streamlit, SQLite, n8n, ffmpeg, OpenCV

EXPERIENCE

L&T Finance, Data Analyst Intern Jun 2025 – Aug 2025

- Validated Nostradamus, an 8-model MLOps platform on GCP and Kubeflow spanning EWS, Banking, Self-Cure and Collections workflows, ahead of its production release.
- Caught an output-formatting defect in the EWS model that left its output column blank downstream, a failure that surfaces as missing data rather than as an error, and drove root-cause analysis with the developers and MLOps engineers who owned the pipeline.
- Wrote SQL verification across pipeline stages: data-integrity and row-level checks that separated “the pipeline ran” from “the pipeline produced the right numbers”.
- Ran end-to-end pipeline validation and UAT in Python and SQL, covering test planning, defect tracking and deployment-readiness review.

E-Cell, IIT Bombay, Campus Executive Aug 2023 – Feb 2024

- Promoted entrepreneurship programs, workshops and competitions, and coordinated outreach, event logistics and mentor coordination across the student community.

PROJECTS

moneytrail: a bank-statement ledger that provably balances github.com/veer0608/moneytrail

Python · pdfplumber · SQLite · pytest · GitHub Actions

- Built a local-first tool that turns bank statements into a ledger. It starts with a **reconciliation gate** rather than a categoriser: every parsed row is checked against arithmetic the bank already published, walking the running-balance column and confirming that opening plus credits minus debits equals closing. A dropped row surfaces as a line number and a rupee delta instead of a quietly wrong category total.
- Stored money as an integer count of paise throughout, with no floats anywhere in the codebase, because binary floats cannot represent exact rupee amounts. Values are formatted as rupees only at the edge.
- Parses CSV, TSV, spreadsheets and real bank PDFs, including password-protected files and borderless tables, where columns are recovered from where words sit under the header and wrapped narrations are stitched back together. Scanned PDFs are rejected rather than guessed at, and card statements reconcile against the issuer's summary box instead of a balance chain.
- Verified against a real bank export, not only fixtures: 225 transactions across three months reconcile to the paisa, and against the closing balance the bank prints in its own summary block. First contact with real data surfaced four defects no synthetic fixture could have found, including amounts masked with asterisks and an IFSC code read as a reference number. Each is now a test.
- 499 tests, CI green on Python 3.11 and 3.12. Also shipped as a paid n8n template: extraction is deterministic, the model only assigns column roles behind a structured-output schema, and arithmetic checks the result. In a live run the model mislabelled a salary row and the row-by-row walk caught it, while an opening-versus-closing check alone would have passed it.

AgentOps: an AI support engineer that takes gated actions github.com/veer0608/agentops

Python · FastAPI · LangGraph · SQLAlchemy · Alembic

- Built an AI support agent that takes real actions (create and update tickets, escalate) behind a policy gate. Each action is auto-executed, routed to a human review queue, or blocked, based on business rules and confidence thresholds. Writes are idempotent and recorded in an append-only audit log.
- Made the model provider-agnostic, with OpenAI, Anthropic and Google Gemini swappable through one config value, and implemented a second agent on LangGraph behind the same interface. The eval harness shows both implementations score identically.

- Wrote a scenario-based evaluation harness scoring six dimensions (tool-selection precision, recall and F1, task-success, escalation accuracy, citation-grounding, cost, latency) and persisted every run for before-and-after tracking. FastAPI, SQLAlchemy and Alembic on SQLite in development and Postgres in production; 36 automated tests.
- Ran the suite eight times against a live model rather than once. Tool-selection F1 rose from a 0.60 baseline to 0.78 on the mean, and citation-grounding held at 1.000 with zero deviation across 80 scenario-runs. Run-to-run spread was 0.11, wider than the effect being measured, so the writeup reports a mean and a spread rather than whichever single pass looked best.

CiteRAG: retrieval-augmented QA with a golden-set eval github.com/veer0608/citerag

Python · FastAPI · SQLite-vec · bge embeddings and cross-encoder

- Built a retrieval-augmented QA system over messy 10-K PDFs, where every answer is grounded in cited source chunks with page numbers.
- Created a golden-set eval harness (recall@k, MRR) and raised recall@5 from 0.37 to 0.77 against a measured ceiling of 0.85, by reworking the chunking strategy and adding a cross-encoder re-ranker. The largest single gain came from fixing PDF text extraction, not from retrieval design.
- Local embeddings and re-ranker on SQLite-vec, so the retrieval path and the whole eval harness run without a paid API. 77 tests, CI on every push.

bhavcopy-pipeline: a daily market-data pipeline github.com/veer0608/bhavcopy-pipeline

Python · DuckDB · dbt · pytest · GitHub Actions

- Built a daily pipeline for NSE and BSE end-of-day prices with no model in it: each file is landed as an immutable gzip, loaded into DuckDB, then modelled and tested with dbt, on a GitHub Actions schedule. 563,111 rows across 66 trading days per exchange.
- Made the load **idempotent and restatable**, and tested both: re-running a day changes nothing, an incremental build equals a full refresh, and a republished file replaces its day whole. 18 dbt data tests.

schemablind: a schema-blind SQL agent github.com/veer0608/schemablind

Python · SQLite · BIRD benchmark · pytest

- Built a SQL agent that is given no schema. It explores a database it has never seen through four read-only verbs (list tables, describe, sample rows, run query) and composes its own answer, recovering from its own failed queries as it goes.
- Scored on BIRD execution accuracy: the agent's rows against the reference query's rows on the same database, with no judge model deciding whether the SQL looks right. 40% of questions are held back as a test split assigned by hash of the question text, so a prompt tuned on failures cannot choose what it is marked on.
- Tested the harness before trusting it: an oracle solver must score 100% and a mute solver 0%, asserted in CI on every push. Model-composed SQL runs behind four independent sandbox layers, and CI asserts the database is byte-identical after attempted writes. A run that exhausts its token budget is reported as unscored rather than as a partial number. 119 tests.

vidsmith: script to narrated, captioned video github.com/veer0608/vidsmith

Python · edge-tts · ffmpeg · Gemini vision

- Built a pipeline that turns a markdown script into a delivery-ready mp4: neural narration, a shot per scene, burned-in word-timed captions, music ducked under the voice, plus an .srt file and a draft title, description and chapters.
- Caption and cut timings come from the TTS engine's own word-boundary events rather than from transcribing the audio it had just produced, so there is no Whisper dependency and nothing to drift. Shots cut on the sentence boundaries those same events report, and candidate stock clips are re-ranked by a vision model against the narration line, because stock search ranks by popularity rather than by what is in frame.

AgencyDesk: a multi-tenant SaaS platform github.com/veer0608/agencydesk

React · FastAPI · PostgreSQL · Docker

- Built a multi-tenant client and project management platform where one deployment serves many isolated agencies, each with its own clients, projects and client portal.
- Enforced tenant isolation and internal-versus-client visibility with PostgreSQL row-level security and composite foreign keys, so cross-tenant access is impossible even if application code is wrong, and the API connects as an unprivileged database role. 70 automated tests cover cross-tenant isolation and role-based visibility. Containerized with Docker Compose, with GitHub Actions CI on every push.

EDUCATION

Madhav Institute of Technology & Science, Gwalior, B.Tech, Computer Science & Design, 2022 – 2026

Shanti Niketan HS School, Vidisha, Class 12 (2021) • Trinity Convent Sr. Sec. School, Vidisha, Class 10 (2019)

ALSO ON GITHUB

finquery-eval-lab, schema-constrained text-to-query for financial questions. The model only translates language into a validated query; deterministic code does the arithmetic over integer paise and returns the evidence rows. Explicit refusals, model-versus-baseline evals, cost and latency tracked. · **geojit-report-app**, turns a financial context document into an auto-filled equity-research PDF (FastAPI and ReportLab), with rule-based extraction as a fallback so generation never fails. · **data-analysis-agent**, automated EDA behind an API: profile, server-rendered charts, IsolationForest and Z-score outliers, downloadable PDF report. · **travel-admin-organizer**, an Activepieces agent that turns Gmail booking confirmations into one itinerary, buffered calendar events and an expense sheet per trip.

CERTIFICATIONS & ACTIVITIES

Affective Computing, NPTEL (Jan–Apr 2024) • Digital Design with Verilog, NPTEL (Jan–Apr 2025) • Rational Thinking, Udemy R&D Coordinator, Startup Cell, MITS (Jan 2023 – Jun 2026) • Core Team Member, Digital Learning Group, MITS (Feb 2023 – Jul 2024) • State-level swimmer

Languages: English, Hindi